

WebVR

Chris Swan @cpswan
CTO, GIS at CSC

and another former RN WEO
(there is no conspiracy)

VR might seem complex and expensive now

1 3XS Z170 Performance Virtual Reality



Specification

Overview

- Intel Core i5 6600K Quad Core professionally overclocked up to 4.6GHz
- Asus Z170-K, Intel Z170 Chipset
- 8GB Corsair Vengeance DDR4 3000MHz
- 6GB EVGA GTX 1060 SC
- 250GB Samsung 750 EVO & 1TB HDD
- Microsoft Windows 10 64bit
- 3 Year Premium Warranty

✓ Award Winning

✓ Next Day Delivery

✓ Priority Build Option

✓ Professional Build

System Powered By



Price from

£866.64 ex VAT

£1039.97 inc VAT

Order now for delivery between
Tuesday 20 Sep - Wednesday 21 Sep

Configure

>

9 **LN70472** Compare ☐



HOT SELLER HTC Vive Virtual Reality Headset, with 2 Wireless Controllers + Movement Sensors, for the Ultimate VR Gaming Experience

High End 

£632.50 ex VAT

£759.00 inc VAT

✓ In Stock

 **BUY**

But all you need is a mobile phone and
'cardboard'



Gankarii - DIY Google Cardboard
Headset 3D Virtual Reality VR
Goggles for Android Smart Phones
iPhone 6s
by Gankarii

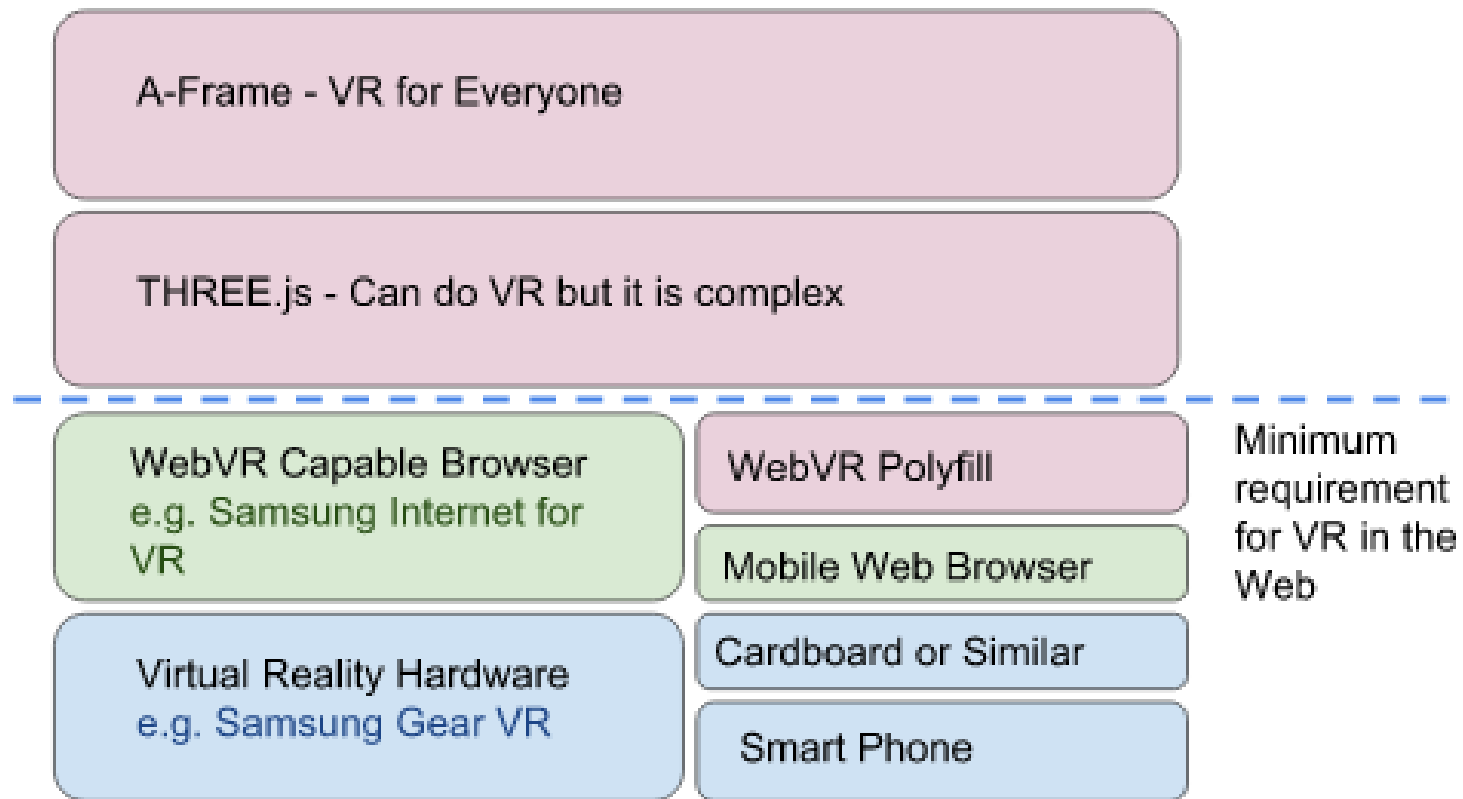
£3.96 ~~£5.30~~  **Prime**

Get it by **Tomorrow, Sep 16**

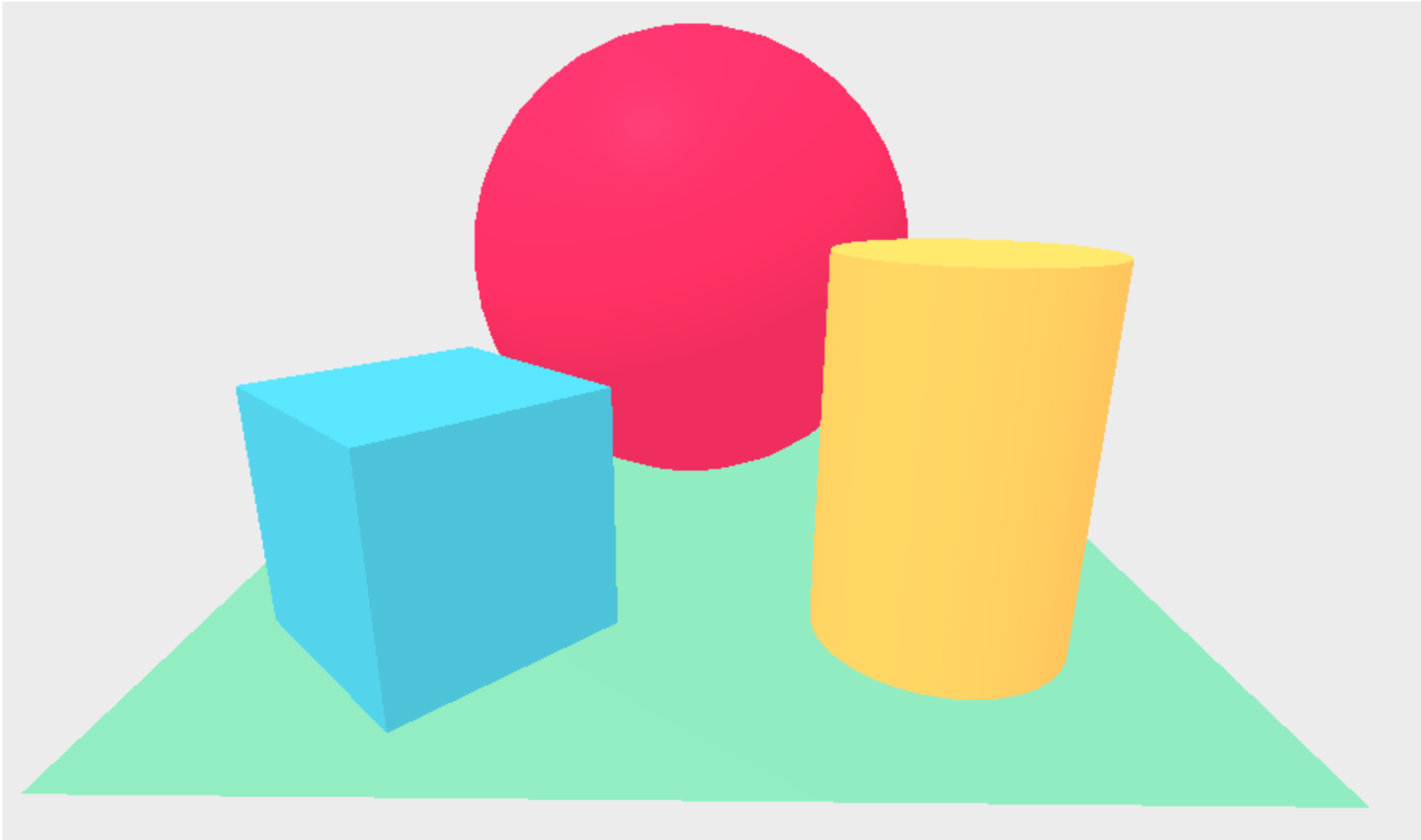
 32

And a development platform

<https://hub.docker.com/r/cpswan/webvr/> (or is.gd/WebVR)



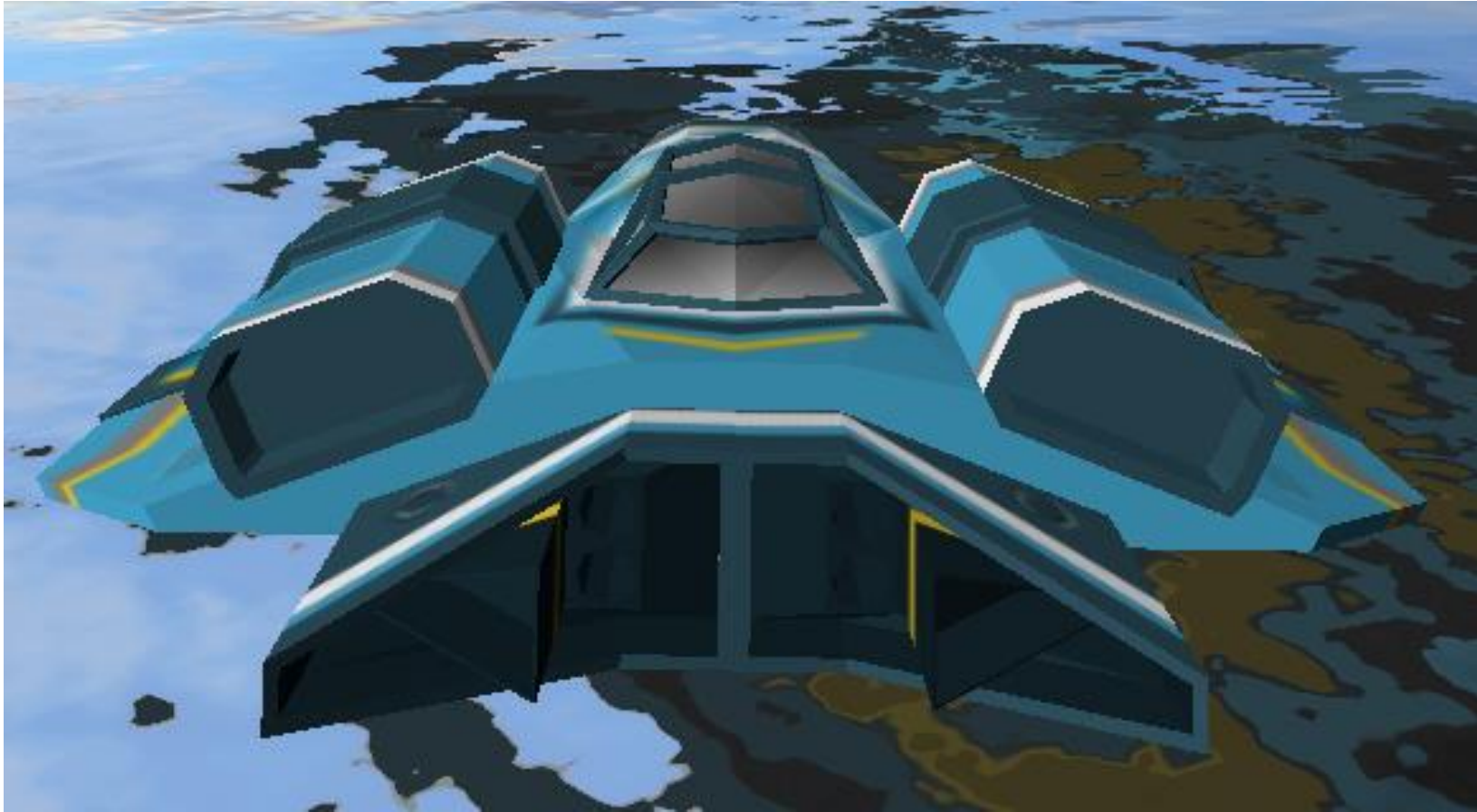
Basic Demo



Basic Demo code

```
<a-scene>  
  <a-sphere position="0 1.25 -1" radius="1.25" color="#EF2D5E"></a-sphere>  
  <a-cube position="-1 0.5 1" rotation="0 45 0" width="1" height="1" depth="1" color="#4CC3D9"></a-cube>  
  <a-cylinder position="1 0.75 1" radius="0.5" height="1.5" color="#FFC65D"></a-cylinder>  
  <a-plane rotation="-90 0 0" width="4" height="4" color="#7BC8A4"></a-plane>  
  <a-sky color="#ECECEC"></a-sky>  
</a-scene>
```

Track Demo



Or swap the spaceship for a dinosaur



Trex code

```
<a-scene inspector="">
  <!-- Assets -->
  <a-assets>
    <a-asset-item id="Trex-obj" src="a-frame-assets/Trex_OBJ/Trex.obj"></a-asset-item>
    <a-asset-item id="Trex-mtl" src="a-frame-assets/Trex_OBJ/Trex.mtl"></a-asset-item>
    <a-asset-item id="race-track-obj" src="a-frame-assets/race-track/race-track.obj"></a-asset-item>
    <a-asset-item id="race-track-mtl" src="a-frame-assets/race-track/race-track.mtl"></a-asset-item>
    
    <!-- Sky is free sample sky from cgskies, buy one for commercial use -->
    
  </a-assets>
  <!-- CAMERA -->
  <a-entity look-at="#trex" ada-follow="target: #trex-camera-target;">
    <a-entity position="0 6 0" rotation="0 180 0">
      <!-- Disable the default wasd controls we are using those to control the ship -->
      <a-camera wasd-controls="enabled: false;"></a-camera>
    </a-entity>
  </a-entity>
  <!-- TRex (from turbosquid) -->
  <a-entity ada-ship-controller="easing: 5; acceleration: 100;" rotation="0 90 0">
    <a-entity id="trex-camera-target" position="0 0 15"></a-entity>
    <a-obj-model src="#Trex-obj" mtl="#Trex-mtl" rotation="0 180 0" position="0 1 0" scale="0.5 0.5 0.5" id="trex"></a-obj-model>
  </a-entity>
  <!-- ENVIRONMENT -->
  <a-entity light="color: #FFFFFF; intensity: 0.3; type: ambient;"></a-entity>
  <!-- SKY SHADER -->
  <!--<a-ada-sky control="#sun" inclination="0.49">
    <a-entity light="color: #FFFFFF; intensity: 1.5" id="sun"></a-entity>
  </a-ada-sky>-->
  <!-- Prerendered for performance -->
  <a-sky src="#cgsky" position="0 -1 0" rotation="0 -90 0">
    <a-entity light="color: #FFFFFF; intensity: 1.5" position="0 1 50"></a-entity>
  </a-sky>
  <a-ada-ocean position="0 0 0" src="#water-normal" width="1000" depth="1000" light="#sun"></a-ada-ocean>
  <!-- TRACK -->
  <a-curve id="track" type="CatmullRom">
    <a-curve-point position="0 0 0"></a-curve-point>
    <a-curve-point position="1 0 0"></a-curve-point>
    <a-curve-point position="0 0 1"></a-curve-point>
  </a-curve>
  <a-draw-curve curve="#track" material="shader: line; color: red;"></a-draw-curve>
  <a-clone-along-curve a-obj-model="src: #race-track-obj; mtl: #race-track-mtl;" curve="#track" spacing="5"></a-clone-along-curve>
</a-scene>
```

I only made a Dockerfile

[Ada Rose Edwards](#) (@Lady_Ada_King), and the amazing A-Frame community did all of the hard work.

Check out her presentation at <https://ada.is/getting-started-with-webvr/> (or is.gd/AdaVR).

